

CND-IDS: Continual Novelty Detection for Intrusion Detection Systems

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Abstract—Intrusion detection systems (IDS) play a crucial role in IoT and network security by monitoring system data and alerting to suspicious activities. Machine learning (ML) has emerged as a promising solution for IDS, offering highly accurate intrusion detection. However, ML-IDS solutions often overlook two critical aspects needed to build reliable systems: continually changing data streams and a lack of attack labels. Streaming network traffic and associated cyber attacks are continually changing, which can degrade the performance of deployed ML models. Labeling attack data, such as zero-day attacks, in real-world intrusion scenarios may not be feasible, making the use of ML solutions that do not rely on attack labels necessary. To address both these challenges, we propose CND-IDS, a continual novelty detection IDS framework which consists of (i) a learning-based feature extractor that continuously updates new feature representations of the system data, and (ii) a novelty detector that identifies new cyber attacks by leveraging principal component analysis (PCA) reconstruction. Our results on realistic intrusion datasets show that CND-IDS achieves up to $6.1\times$ F-score improvement, and up to $6.5\times$ improved forward transfer over the SOTA unsupervised continual learning algorithm. Our code is available at <https://github.com/Sean-Fuhrman/CND-IDS>.

I. INTRODUCTION

In today's digital landscape, cybersecurity is essential for safeguarding sensitive data and maintaining trust in digital systems. With cyber threats becoming more sophisticated, organizations must adopt thorough security measures to protect against breaches and unauthorized access [1]. A critical part of such security measures are intrusion detection systems (IDS) which identify malicious network traffic and computer usage [2]. Recently, Machine Learning (ML) has become a popular IDS solution due to its great attack detection performance and less requirement for human knowledge [3]. However, state-of-the-art (SOTA) ML-IDS solutions often fail to adequately address the challenges posed by (i) the evolving nature of cyber attacks and (ii) the lack of attack labels.

A University of Maryland survey predicts that a new cyber attack happens somewhere on the Internet every 39 seconds [4], underscoring the need for dynamic IDS. However, most SOTA ML-IDS approaches are static (offline) and cannot easily adapt their behavior over time [5]. ML models, when taught new knowledge, *catastrophically forget* previously known information. To efficiently address this problem, continual learning (CL) is proposed which is capable of learning from an infinite stream of data [6]. An experience is defined as a shift

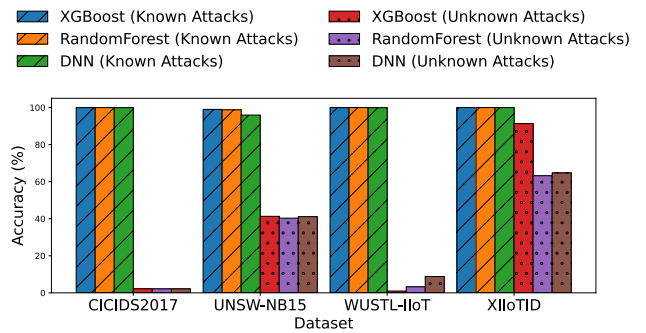


Fig. 1: State of the art machine learning intrusion detection performance on known/unknown attacks

in the data stream distribution, and CL models are designed to function well across multiple experiences. This capability is currently essential in cybersecurity, where each new attack type can be considered as a new experience [7]. To illustrate, an intrusion detection model trained for a group of attacks (experiences) such as *worm*, *shellcode*, and *exploits* should be able to identify new types of attacks (experiences) such as *backdoor* and *fuzzers* [8]. Some studies have adopted CL for intrusion detection [7]–[12], but they rely on labeled attack data, which may not always be feasible to obtain. This causes the ML model to overfit to common attacks in the dataset, resulting in a failure to detect rare and zero-day threats.

SOTA ML-IDS solutions have been shown to have high accuracy on well represented attacks within the data, however have low precision on infrequent attacks [13]. It is extremely difficult for zero-day attacks to be well represented in the data. Figure 1 illustrates the limitations of current ML-IDS methods in handling zero-day attacks, revealing a substantial decline in performance when faced with previously unseen threats. To address this issue, novelty detection (ND)-based ML methods have demonstrated significant promise in detecting both known and zero-day attacks without requiring a labeled dataset [14]. These ND solutions can effectively identify attacks by analyzing purely normal data, provided that the attacks exhibit substantial deviations from normal behavior [14]. However, it has been shown that ND algorithms such as Isolation Forest and Local Outlier Factor result in lower prediction

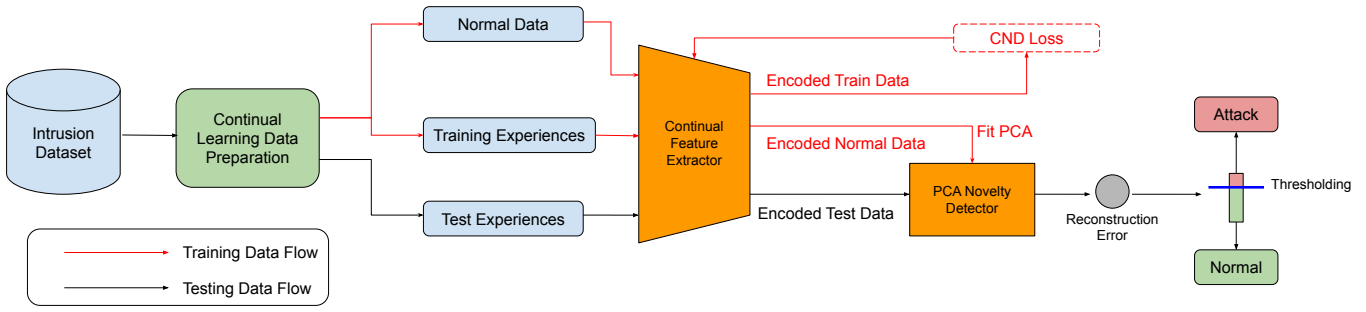


Fig. 2: Continual Novelty Detection IDS (CND-IDS) Framework

performance under changing environments, where they suffer from issues like *catastrophic forgetting* [15]. This issue can be addressed by leveraging continual learning strategies, hence motivating continual novelty detection methods for IDS.

To address the challenges of evolving data streams and lack of supervision, we propose *CND-IDS*, a continual novelty detection approach for ML-based intrusion detection (Fig. 2). To the best of our knowledge, we are the first to propose continual ND for intrusion detection. CND-IDS consists of two critical components: a continual feature extractor (CFE) and a novelty detector based on principal component analysis (PCA) reconstruction. We design a novel continual ND loss to learn representations of the data stream, enabling the CFE to update its representations in a streaming environment with zero attack labels. PCA reconstruction error is then used to classify test samples as either attack or normal data based on thresholding. By fitting PCA on normal data, CND-IDS is trained without any labeled attack data, enabling it to detect previously unknown zero-day attacks. Our results show that on realistic intrusion datasets, CND-IDS achieves up to $6.1 \times$ F-score improvement over the SOTA unsupervised continual learning algorithm.

II. RELATED WORK

Intrusion Detection Systems: Intrusion Detection Systems (IDS) fall into two main categories: Signature-based and Anomaly-based IDS. Signature-based IDS identify attacks by matching network data to known attack patterns. In contrast, Anomaly-based IDS use machine learning methods to analyze statistical properties of network data [16]. Machine learning-based IDS (ML-IDS), a popular form of Anomaly-based IDS, use historical data to train models, enhancing accuracy and reducing the need for expert knowledge [17]. The literature proposes various ML models, including supervised models such as support vector machine, random forest, deep neural networks [18], and unsupervised models like isolation forest, autoencoder, and principal component analysis [14].

Continual Learning for Intrusion Detection: Current ML-IDS solutions are often built on the premise that all data are sampled *i.i.d.* (independent and identically distributed) [5], making them vulnerable to performance degradation when data distributions shift. Continual Learning (CL) seeks to train ML

models to acquire new information while minimizing the loss of previously learned knowledge. Prasath et al. [9] highlight that intrusion datasets exhibit these shifting distributions, underscoring the need for CL-capable IDS that can autonomously adapt to the evolving characteristics of the data. Few studies have recently adopted CL for intrusion detection. Kozik et al. [7] extend efficient lifelong learning algorithm [19] to detect cyber attacks and address the data imbalance problem. Kumar et al. [20] applied Elastic Weight Consolidation and Gradient Episodic Memory CL algorithms. In a follow-up study [10], they extend a memory-based CL method and proposed a novel approach (perturbation assistance for parameter approximation), improving CL scalability. Nevertheless, all of these studies leverage labeled attack data to devise their CL solutions, which may not be feasible in IDS due to the difficulty and expense of obtaining such labeled data.

Continual Novelty Detection: Continual Novelty Detection (CND) primarily addresses detecting distribution shifts after ML model deployment. To illustrate, Kim et al. [21] highlight how ND combined with CL is necessary for realistic class incremental learning. Aljundi et al. [22] investigate CND for detecting data outside the current model’s knowledge. Finally, Rios et al. [23] propose a novel CND framework for detecting new data and incorporating it into the current ML model’s knowledge. These studies focus on using ND to identify data outside the training set and integrate it into the ML model via CL strategies. However, Faber et al. [15] demonstrate the performance decline in ND-algorithms themselves under continual learning settings. Given the constantly changing nature of attacks, it is necessary for ML-IDS systems to continually adapt without labels, which motivates our work. To the best of our knowledge, we are the first to propose continual novelty detection for ML-based intrusion detection.

III. CND-IDS FRAMEWORK

Fig. 2 illustrates our continual novelty detection-based intrusion detection (CND-IDS) framework. We first apply a continual learning data preparation phase, where we divide the intrusion data into normal data, training experiences, and test experiences. Our framework consists of two main components: a continual feature extractor (CFE) and principal component analysis (PCA)-based novelty detector. The CFE creates a

representation of the data that is used to distinguish normal and attack data by the novelty detector. The CFE allows the model to learn new feature representations overtime, so that it can easily adapt as new attacks are introduced through the data stream. CFE is trained through our continual novelty detection loss function, to continually learn from the input stream without forgetting. The next component, PCA reconstruction, provides an anomaly score on the feature output from the feature extractor. PCA is trained purely on normal data, therefore it can generalize well to unseen attacks. Finally we apply a threshold, τ , such that if the anomaly score is greater than τ , we classify that data as an attack. Overall, our learning framework does not require any labeled attack data.

A. Continual Learning (CL) Data Preparation

We assume a given dataset contains attack data A , normal (benign) data N , and various attack types C . To begin our data preparation, we remove 10% of the normal data, and save this as clean normal data N_c to train PCA novelty detector. For the remaining data, we split it across an arbitrary number of experiences m . We create a set of experiences E , such that each individual experience is $(E_0, E_1, \dots, E_m)$. Each experience contains a portion of the remaining normal data, of size $\frac{0.9*|N|}{m}$. The attacks are distributed such that each experience contains $\frac{|C|}{m}$ different attack classes. Therefore, each experience contains some number of *unique* attacks to that experience. This allows us to test on zero-day attacks through future experiences, and seen attacks through current or past experiences. Each experience E_i is split into training and testing set. The training split contains only X_{train} data points, meaning it does not include any labels. The test data contains $\{X_{test}, Y_{test}\}$ where $Y_{test} \in \{0, 1\}$, where 0 means normal, and 1 means attack data. For each experience E_i , we utilize the training set X_{train}^i and N_c to train CND-IDS. We then evaluate the model using the test points $\{X_{test}, Y_{test}\}$. This setup is designed to represent a realistic intrusion detection scenario, where attacks are entirely unknown before deployment into a continual learning environment. In our framework, the only known data is N , while all attack data is unlabeled. With this data setup, we can assess model's performance across various attack scenarios. Specifically, we can simulate the model's response to zero-day attacks as well as known attacks.

B. Proposed Algorithm

Algorithm 1 depicts the proposed CND-IDS algorithm, including the training and test steps. The training flow follows three main steps: (i) train the CFE, (ii) encode N_c into features, and (iii) train PCA-based novelty detector on the encoded N_c data. Each of these steps is performed at every training experience. For step (i), the CFE is trained on X_{train} using the loss function from Equation 1. For step (ii), the subset of normal data N_c is encoded by passing it through the trained CFE, where we encode the entire set. Ultimately, in step (iii), we fit the PCA novelty detection model to this encoded normal data. This completes the CND-IDS training, and we then proceed to evaluate on the test experiences.

Algorithm 1 CND-IDS Algorithm (Training and Test)

Require: E - Experience Set; N_c - Subset of Clean Normal Data; CFE - Continual Feature Extractor; ND - PCA Novelty Detector

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1: for  $i \leftarrow 1$  to  $|E|$  do
2:   Get  $X_{train}$  from experience data  $E_i$ 
3:   Fit  $CFE$  to training data  $X_{train}$ 
4:   Encode  $N_c$  by passing it through  $CFE$ 
5:   Fit  $ND$  to encoded  $N_c$ 
6:   Get  $X_{test}, Y_{test}$  from all experiences within  $E$ 
7:   Encode  $X_{test}$  by passing it through  $CFE$ 
8:   Use  $ND$  to compute anomaly score  $S_{test}$  on test data
9:   Compute threshold  $\tau$  based on thresholding method
10:  Compute predictions such that  $Y_{pred} = (S_{test} > \tau)$ 
11:  Compute evaluation metrics based on  $Y_{pred}, Y_{test}$ 
12: end for

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At the end of each training experience, we evaluate on the test set of all experiences. Therefore, we evaluate model performance on unseen attacks (future experiences) and seen attacks (past experiences). When evaluating a batch of testing data, we encode all points from X_{test} with CFE. Then use PCA reconstruction to get an anomaly score S_{test} for each point based on the reconstruction loss. Next, we leverage widely used Best-F [24] to select a threshold τ . Ultimately, all anomaly scores greater than τ are classified as attack while all scores less than or equal to τ are classified as normal, resulting in the CND-IDS predictions Y_{pred} . We ultimately compute our evaluation metrics by comparing Y_{pred} with the true labels Y_{test} .

C. Continual Feature Extractor (CFE)

CFE is an autoencoder (AE)-based model that leverages Multi-Layer Perceptron (MLP) as both the encoder and decoder. Our continual novelty detection loss (L_{CND}) consists of a novel cluster separation loss (L_{CS}), a reconstruction loss (L_R), and a continual learning loss (L_{CL}):

$$L_{CND} = L_{CS} + \lambda_R L_R + \lambda_{CL} L_{CL} \quad (1)$$

where $\lambda_R \in [0, 1]$ and $\lambda_{CL} \in [0, 1]$ are hyper-parameters controlling the strength of the reconstruction loss and the continual learning loss respectively.

Cluster Separation Loss: We design a novel cluster separation loss L_{CS} to enhance the performance of our PCA-based novelty detector by increasing the separation between normal and anomalous data points in feature space. To achieve this, we leverage the clean normal data N_c (which is also used to train the PCA model). Using K-Means clustering [25], we identify which points in X_{train} are most similar to N_c and then push these points apart in feature space. Specifically, L_{CS} leverages K-Means clustering to assign pseudo-labels to all points in X_{train} . Then it utilizes triplet margin loss [26] to maximize the Euclidean distance between the different pseudo labels. Calculating the L_{CS} pseudo-labels involves the following steps: 1) Fit K-Means clustering to all points in

X_{train} ; 2) Find the cluster labels of all N_c data points; 3) Create the set of "normal data" clusters CL_N , where each cluster contains at least one point from N_c ; 4) Assign class 0 to points in X_{train} if their associated cluster is in CL_N , and class 1 otherwise. In summary, after fitting K-Means to X_{train} , we identify the clusters associated with N_c . If a cluster contains any point from N_c , all points in that cluster are assigned to class 0; otherwise, they are assigned to class 1. This effectively splits the data into two, where 0 would represent normal pseudo-label, and 1 would represent anomalous pseudo-label.

After assigning the pseudo-labels, the final loss is computed using the widely adopted triplet margin loss [26], defined as:

$$L_{CS} = \max(\Delta_{ap} - \Delta_{an} + m, 0) \quad (2)$$

where Δ_{ap} represents the Euclidean distance between an anchor point and a positive point (a point of the same class), and Δ_{an} is the Euclidean distance between the anchor and a negative point (a point from a different class). The term $m > 0$ is a hyper-parameter that specifies the desired margin between positive and negative points.

Reconstruction Loss: The reconstruction loss ensures that the features embedded by the encoder retain significant information from the original data. We found that this approach helps the PCA extractor better generalize to future and past experiences by forcing the model to learn encodings that align more closely with the original data. The reconstruction loss is the mean squared error (MSE) between the original input and the reconstructed output of the decoder. Let h be the encoded feature of point x . Then $L_R = MSE(\text{decoder}(h), x)$.

Continual Learning Loss: Our solution to catastrophic forgetting is employing a latent-based regularization loss, L_{CL} [27]. This loss ensures that as the model learns new information, it retains knowledge from past experiences. L_{CL} can be formulated as: $L_{CL} = \sum_{i < c}^c MSE(h^c, h^i)$ where c represents the current experience, and h^j is the encoded embedding of input x at experience j . The loss is computed by summing the MSE between the current embedding (h^c) and all previous embeddings h^i where $i < c$. This ensures that the current embedding is still similar to the previous embedding, thereby preventing catastrophic forgetting. To calculate past embedding, we pass the current data point x into a past version of the model. This requires the model to save its state between experiences but does not require it to save any data, which can significantly reduce storage overhead.

D. PCA-based Novelty Detection

Inspired by [23], we create a PCA-based novelty detection approach. Let $\mathbf{h}$ denote the output feature from our feature extractor. We utilize principal component analysis transformation $\mathbf{T}$ to map the input feature from a high dimensional space to low dimensional space $\mathbf{T} : \mathbf{h} \rightarrow \mathbf{l}$. We then utilize the feature reconstruction error (FRE) as the anomaly score: $FRE = \|\mathbf{h} - \mathbf{T}^{-1}(\mathbf{l})\|^2$, where $\mathbf{T}^{-1}$ is the inverse PCA transform. The anomaly score is therefore the reconstruction loss from this method. We train the PCA transformation on the subset of normal data N_c after it is encoded by the CFE.

IV. EXPERIMENTAL ANALYSIS

A. Experimental Setup

Datasets: Table I provides a detailed breakdown of the SOTA intrusion datasets utilized in our study. For IIoT intrusion, we use IIoT datasets X-IIoTID [28] and WUSTL-IIoT [29]. We also include commonly used network intrusion datasets CICIDS2017 [30] and UNSW-NB15 [31]. For X-IIoTID [28], CICIDS2017 [30], and UNSW-NB15 [31], we split the data across five experiences such that each experience contains two to four attacks. For WUSTL-IIoT [29], we split the data across four experiences such that each experience contains one attack. We perform this data split to simulate an evolving data stream with emerging cyber attacks over time where each experience contains different attacks.

TABLE I: Selected Intrusion Datasets

Dataset	Size	Normal Data	Attack Data	Attack Types
X-IIoTID [28]	820,502	421,417	399,417	18
WUSTL-IIoT [29]	1,194,464	1,107,448	87,016	4
CICIDS2017 [30]	2,830,743	2,273,097	557,646	15
UNSW-NB15 [31]	257,673	164,673	93,000	10

Baselines: We evaluate our algorithm against two SOTA unsupervised continual learning (UCL) algorithms: the Autonomous Deep Clustering Network (**ADCN**) [27], and an autoencoder paired with K-Means clustering. The autoencoder K-Means model is combined with Learning without Forgetting [32] continual learning loss; we refer to this model as **LwF**. Note that both **ADCN** and **LwF** require a small amount of labeled normal and attack data to perform classification. We also compare our approach against SOTA ND methods: local outlier factor (**LOF**) [15], one-class support vector machine (**OC-SVM**) [15], principal component analysis (**PCA**) [23], and Deep Isolation Forest (**DIF**) [33]. Since these ND models cannot be retrained on unlabeled contaminated data, continual learning is not feasible for these methods.

Evaluation Metrics: To evaluate the model performance, we report F_1 score. Since there is a class imbalance within these datasets, to simulate real world IDS, F_1 score gives an accurate idea on attack detection. For the continual learning methods, we evaluate their performance at the end of each training experience on all experience test sets. This generates a matrix of F_1 score results R_{ij} such that i is the current training experience, and j is the testing experience. To summarize this matrix of results, we report widely used CL metrics [34]: average F_1 score on current experience (AVG), forward transfer (FwdTrans), and backward transfer (BwdTrans). For a matrix R_{ij} with m total experiences, our metrics are formulated as follows: $AVG_{F_1} = \frac{\sum_{i=j} R_{ij}}{m}$; $FwdTrans_{F_1} = \frac{\sum_{j>i} R_{ij}}{m*(m-1)}$; $BwdTrans_{F_1} = \frac{\sum_{i=1}^m R_{mi} - R_{ii}}{\frac{m*(m-1)}{2}}$. AVG is the average performance on the current test experience at every point of training. FwdTrans is the average performance on "future" experiences, which simulates performance on zero-day attacks. Finally, BwdTrans is the average change in performance of "past" test experiences at a "future" point of training. A negative BwdTrans indicates catastrophic forgetting, whereas a positive

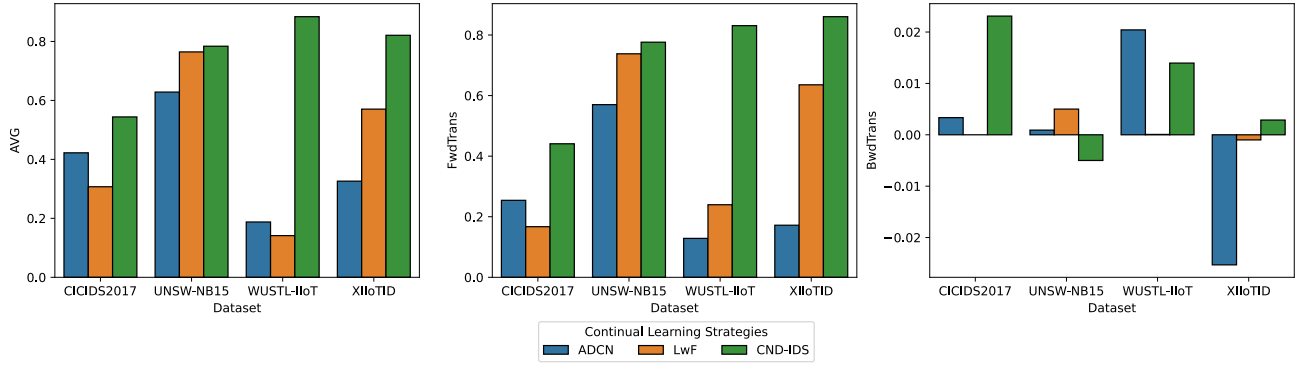


Fig. 3: Continual learning metric results of ADCN [27], LwF [32], and CND-IDS

BwdTrans indicates the model actually improved performance on past experiences after learning a future experience. Overall, AVG measures seen attacks, FwdTrans measures zero-day attacks, and BwdTrans measures forgetting. For all metrics, a higher positive result indicates a better performance.

We also report the threshold-free metric Precision-Recall Area Under the Curve (PR-AUC) [35]. Since CND-IDS requires selecting a threshold, PR-AUC allows us to assess model performance independently of the threshold. We choose PR-AUC over Receiver Operating Characteristic Area Under the Curve (ROC-AUC) because ROC-AUC can give misleadingly high results in the presence of class imbalance [35].

Hyperparameters: We utilize *elbow method* [36] for determining the number of clusters K . We set λ_R and λ_{CL} to 0.1, and for m we use 2 after careful experimentation. For the AE modules of CND-IDS, we use 4-layer MLP with 256 neurons in the hidden layers. We train it using Adam optimizer [37] with a learning rate of 0.001. For PCA, we use the explained variance method and set it to 95% [23].

Hardware: We run our experiments on NVIDIA GeForce RTX 3090 GPU, with a AMD EPYC 7343 16-Core processor.

B. Results

Continual Learning Comparison: Fig. 3 presents the results of our approach CND-IDS compared with ADCN [27] and LwF [32]. CND-IDS shows the best performance on both seen (AVG) and unseen (FwdTrans) attacks across all datasets. CND-IDS also has the highest BwdTrans on all except one dataset (UNSW-NB15). The average BwdTrans of CND-IDS (0.87%) is higher than the average BwdTrans of both ADCN (-0.06%) and LwF (0.09%). Notably, the BwdTrans of CND-IDS is positive for three datasets. Indicating past experiences actually improve after training on future experiences for these datasets. Given the high FwdTrans as well, our approach finds features that generalize well to future experiences.

Table II shows the improvement of CND-IDS over the UCL baselines on all datasets. Bold and underlined cases indicate the best and the second best improvements with respect to each metric, respectively. These improvements were calculated by comparing the performance of CND-IDS to the baselines, where the improvement values represent the

TABLE II: CND-IDS Improvement over UCL Baselines

Baseline	Dataset	AVG	FwdTrans
ADCN [27]	X-IIoTID	2.02×	5.00×
	WUSTL-IIoT	4.50×	6.47×
	CICIDS2017	1.37×	1.73×
	UNSW-NB15	1.29×	1.44×
LwF [32]	X-IIoTID	1.46×	1.35×
	WUSTL-IIoT	6.11×	3.47×
	CICIDS2017	1.93×	2.64×
	UNSW-NB15	1.11×	1.02×

proportional increase over the baseline performance. We do not include BwdTrans because a proportional increase does not make sense for a metric that can be negative. CND-IDS has up to $4.50\times$ and $6.1\times$ AVG improvement on ADCN and LwF, respectively. In addition, CND-IDS has up to $6.47\times$ and $3.47\times$ FwdTrans improvement on ADCN and LwF. Averaged across all datasets, CND-IDS shows a $1.88\times$ and $1.78\times$ improvement on AVG, and a $2.63\times$ and $1.60\times$ improvement on FwdTrans, compared to ADCN and LwF, respectively.

Overall, these results highlight the benefit of continual ND over UCL methods for IDS. CND-IDS, with its PCA-based novelty detector, excels by effectively harnessing the normal data to identify attacks. A key strength of our approach lies in the assumption that normal data forms a distinct class, while everything else is treated as anomalous. This assumption is particularly well-suited to IDS. In contrast, methods like ADCN and LwF do not make this distinction where they handle both normal and attack data similarly, limiting their ability to fully exploit the inherent structure of the data.

Novelty Detectors Comparison: Fig. 4 compares LOF [15], OC-SVM [15], PCA [23], and DIF [33] with CND-IDS on all datasets. The average F_1 score of the novelty detection methods are compared to the AVG of CND-IDS. It can be seen CND-IDS outperforms all other methods across all datasets. The two best performing methods are DIF and PCA. The average F_1 score improvement across all datasets of CND-IDS is $1.16\times$ and $1.08\times$ over DIF and PCA, respectively. These results highlight the critical role of leveraging information from unsupervised data streams. Unlike these ND algorithms, CND-IDS is capable of continuously learning from this unsupervised data, enabling it to enhance PCA reconstruction over time. By

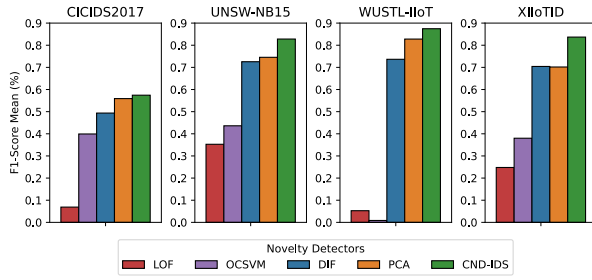


Fig. 4: Average F_1 score on all experiences of CND-IDS and novelty detection methods: LOF, OC-SVM, PCA, DIF

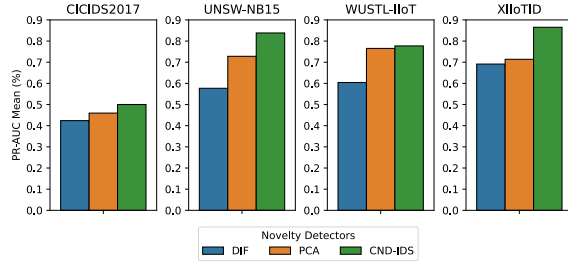


Fig. 5: Thresholding Free Evaluation of CND-IDS

integrating evolving data patterns, CND-IDS not only adapts to new anomalies but also improves its overall detection accuracy, demonstrating a clear advantage in dynamic environments.

Pre-threshold Evaluation: While thresholding plays a crucial role in attack decision-making, evaluating model prediction performance before applying threshold is also important. The UCL algorithms (ADCN [27] and LwF [32]) do not output anomaly scores because they select classes based on the closest labeled cluster. Therefore we compare against the two best ND methods: DIF [33] and PCA [23]. Fig. 5 presents the PR-AUC values of DIF, PCA, and CND-IDS. It can be seen that CND-IDS provides the best threshold free results, which aligns with the threshold-based results presented earlier. The strong performance of CND-IDS in both pre-threshold and threshold-based evaluations demonstrates that the model is robust regardless of the decision threshold.

C. Ablation Study

To demonstrate the impact of our loss function components, we perform an ablation study. Table III shows the results of CND-IDS with each loss function removed to demonstrate their individual effectiveness. Bold and underlined cases indicate the best and the second best performances with respect to each metric, respectively. CND-IDS without reconstruction loss (L_R) and CND-IDS without cluster separation loss (L_{CS}) performs worse in all categories. CND-IDS without both L_R and continual learning loss (L_{CL}) actually performs better AVG but has worse BwdTrans and FwdTrans. AVG does not account for past experiences, so the significantly negative BwdTrans indicates CND-IDS w/o L_R and L_{CL} forgets, and therefore would perform worse on those experiences in the

TABLE III: Ablation Study of CND-IDS Loss Functions

Strategy	AVG	BwdTrans	FwdTrans
CND-IDS	76.92%	0.87%	73.70%
CND-IDS (w/o L_{CS})	66.23%	0.09%	70.26%
CND-IDS (w/o L_R)	72.86%	-5.44%	67.82%
CND-IDS (w/o L_R and L_{CL})	<u>79.92%</u>	-11.26%	<u>71.01%</u>

TABLE IV: Average inference time (in ms) per test sample

Strategy	CND-IDS	ADCN	LwF	DIF	PCA
Inference Time (ms)	<u>0.0019</u>	0.4061	0.0677	1.0535	0.0018

future. This would make sense as a regularization loss to improve continual learning would slightly decrease performance in non-continual scenario. Overall CND-IDS has the best results when taking every metric category into account. Notably the low BwdTrans and FwdTrans of CND-IDS (w/o L_R) showcases how the reconstruction loss helps CND-IDS generalize better to unseen and past data. This highlights the power of L_R to provide good features for continual learning.

D. Overhead Analysis

Table IV evaluates the inference overhead of CND-IDS compared to ADCN [27], LwF [32], DIF [33], and PCA [23]. CND-IDS offers the fastest inference time among continual learning methods. Out of novelty detection methods, CND-IDS is second only to PCA. We attribute the efficiency of CND-IDS to avoiding the clustering classification used by LwF and ADCN. The difference between CND-IDS and PCA is minimal, only 0.0001 milliseconds slower, due to the additional but lightweight step of encoding the data. Considering that the average median flow duration across datasets is 27.77 milliseconds, the overhead introduced by CND-IDS is negligible in the context of real-time traffic flow.

V. CONCLUSION

IDS are essential for ensuring a robust and comprehensive security strategy. ML methods have been recently adopted for this task due to their great prediction performance. However, current ML-IDS solutions are vulnerable to under-represented attacks and constantly changing environments. Therefore, future IDS must be able to (i) generalize to unseen attacks and (ii) adapt to evolving attacks over time. To address these challenges, we present CND-IDS, a continual novelty detection based IDS algorithm. CND-IDS combines PCA novelty detection with a continual feature extractor. CND-IDS is capable of continually learning with no attack labels, allowing it to detect attacks in a changing data stream. We have demonstrated that on realistic intrusion datasets, CND-IDS is up to $6.1\times$ more effective than SOTA unsupervised CL method for detecting seen attacks and up to $6.5\times$ more effective at detecting future attacks.

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